

Nashville Housing Market Analysis

DATA ANALYTICS PORTFOLIO PROJECT • EXPLORATORY DATA ANALYSIS

\$18.4B

TOTAL REVENUE

~56K

TRANSACTIONS

\$327K

AVG. SALE PRICE

■ Executive Summary

An end-to-end analysis of the Nashville housing ecosystem, identifying a distinct growth cycle peaking in 2015. The market is characterized by heavy centralization in urban centers and a primary reliance on the Single-Family residential sector.

■ Market Dynamics & Temporal Trends

- **The 2015 Peak:** Record-high transaction volumes and valuations.
- **Correction Phase:** Post-2015 data indicates a cooling period with declining transaction velocity.
- **Data Anomalies:** A significant gap in records for 2017-2018 suggests either a systemic reporting failure or an extreme market contraction.

■ Key Segment Insights

Single-Family Dominance

Single-Family properties represent over **50% of total market revenue**. This dominance highlights the segment as the primary driver of regional real estate liquidity.

The "Bedroom Ceiling"

Data shows diminishing marginal returns on bedroom counts. While prices rise with size, the premium levels off after 4-5 bedrooms, suggesting that buyers prioritize location and land value over sheer room count.

■ Geographic Distribution

Market activity is highly concentrated: **83% of all sales** occur within Nashville city limits. This indicates a "Hub and Spoke" model where urban core value significantly outweighs peripheral markets.

■ Data Integrity Assessment

Critical Finding: 46% of the dataset contains missing values in key fields (Acreage, Land Value, Building Value). This highlights a significant opportunity for improving data collection protocols for more accurate predictive modeling.

■ Tech Stack & Skills

SQL (Data Cleaning)

Power BI (Visualization)

DAX (Measures)

Statistical Analysis